STUDY MATERIAL FOR CHAPTER 3.6.

SUMMARY AND TRANSLATION OF CHAPTER 6, FROM THE BOOK "BIOLOGÍA CELULAR BIOMÉDICA", PLASMA MEMBRANE SPECIALIZATIONS AND CELL INTERACTIONS WITH THE ENVIRONMENT

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LEARNING OBJECTIVES:

- **To know the plasma membrane specializations in the different cellular domains (apical, lateral and basal).**
- **To understand the molecular structure of the different plasma membrane specializations.**
- **To relate certain human pathologies with molecular and/or cellular alterations in the plasma membrane specializations.**

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INTRODUCTION

Cells are usually connected to one another or to proteins in the extracellular matrix (ECM) which surround them. Cell interactions with its environment are vital for functioning in a particular tissue. In fact, most cells depend on these interactions for their survival. Thus, if they are unable to adhere to the substrate or to other cells, they end up dying by a process called "**anoikis**" (apoptosis caused by the lack of adhesion). These types of cell-cell and cell-ECM interactions are also key in communication and intercellular signaling. The plasma membrane plays an essential role in these processes, developing different structures that facilitate adhesion, contact with other cells or the interaction with the surrounding medium. In this chapter we describe the plasma membrane specializations and the main extracellular molecules with which they interact.

PLASMA MEMBRANE SPECIALIZATIONS

The plasma membrane may have different specializations **(fig. 6-1)** in either: a) the cells´ free surface (whichever is not in contact with other cells), such as microvilli, cilia and flagella (these last two differentiations are explained in chapter 12), and b) the contact surface with neighboring cells or the ECM, which include interdigitations, basal infoldings and cell-cell or cell-EMC junctions.

CA: Adhesion belt
CI: Cilia
DE: Desmosome
FB: Fibroblast
HD: Hemidesmosome
ID: Interdigitations /lateral infolding
MV: Microvilli
PB: Basal infolding
UC: Communicating junction
UO: Occluding junction

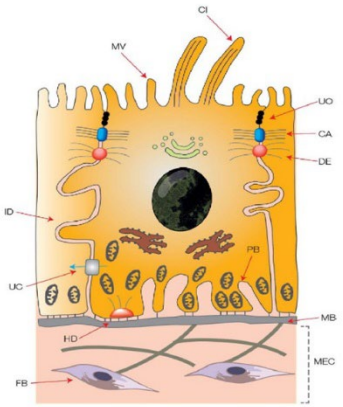


Figure 6.1. Representation of an idealized cell showing the different plasma membrane differentiations studied in this chapter.

The latter can be further classified by their function into:

- Anchoring junctions
- Tight junctions
- Communicating junctions

Microvilli

Microvilli are fingerlike projections in the apical surface of epithelial cells that help increasing the surface area of absorption **(fig. 6-2, A)**. These structures are approximately 1 μm in length and 0.08 μm in width. Inside, they are stabilized by a bundle of 20-30 actin

filaments, linked together by **villin** and **fimbrin**. These membrane differentiations are anchored to the **terminal web**, which contains different filamentous proteins. The microvilli have a highly developed glycocalyx, which can be seen under the optic microscope using saccharide specific stains, such as **PAS** (periodic acid Schiff) **(fig. 6-2,B).**

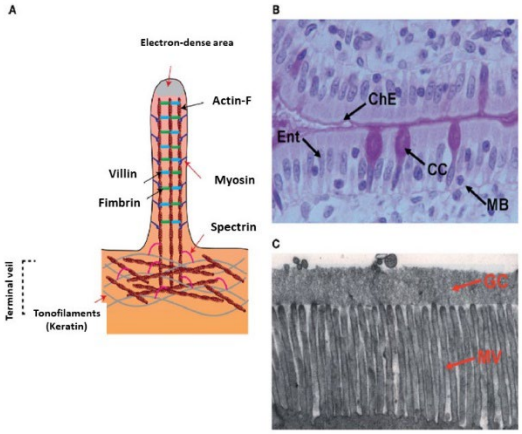


Figure 6.2. Structure of microvilli in enterocytes (Ent). ChE: brush border; CC: Goblet cell. MB: Basal membrane. MV: Microvilli. GC: Glycocalyx.

Microvilli can be scarce in some epithelial cells, but in some others they are very abundant, as is the case for epithelia of the intestine, proximal convoluted tubules and choroid plexus.

In the intestinal epithelium, microvilli **(fig. 6-2, C)** play an essential role in the absorption of nutrients, as they increase the surface area of the membrane, which contains many transporter glycoproteins. Several enzymes, responsible for the final stages of protein and sugar digestion, can be found on the microvilli´ surface. These enzymes can be altered by different causes, resulting in pathologies, either by defect (such as lactose intolerance) or by excessive absorption (such as hemochromatosis). **Lactose intolerance** usually develops during childhood, although its first symptoms may appear in adulthood. In patients with this disease, among other alterations, there is a secondary deficiency of **lactases** (enzymes found in the membrane of the microvilli of enterocytes). The function of these lactases is to split the lactose ingested in foods into its two basic components: glucose and galactose, which will later be absorbed. When there is partial or complete lack of this enzyme, lactose will not be digested, resulting in diarrhea. **Hemochromatosis** is an autosomal recessive disease characterized by an excessive iron build-up in different organs. This disease is characterized by gene mutations, which determines the over-expression in the intestinal microvilli of the DMT-1 transporter, whose function is to absorb iron. Therefore, when an increase in this protein in the epithelial surface occurs (as is the case in hemochromatosis), there is an excessive iron absorption. Because of its high concentration in the patient´s body, iron accumulates in different organs such as the liver heart (causing cardiomyopathy) and hypophysis.

Microfolds: interdigitations and basal infoldings

Microfolds are narrow evaginations/invaginations of the cellular membrane on its contact surface with other cells or with the ECM. They can either be on the lateral surfaces (interdigitations or lateral infoldings), allowing the reinforcement of epithelial cell junctions **(fig. 6-3, A)** or on the epithelial cell´s basal membrane, known as basal infoldings **(fig. 6-3, B)**. These latter ones can be found in the distal convoluted tubules of the kidney, where they are deep and numerous. The cytoplasm within the basal infoldings contains lots of mitochondria. This arrangement relates to the osmotic regulation performed by these cells via ionic transfer, which requires a large energy source. The invaginations increase the surface area of the basal membrane on which the ion transporter proteins are located.

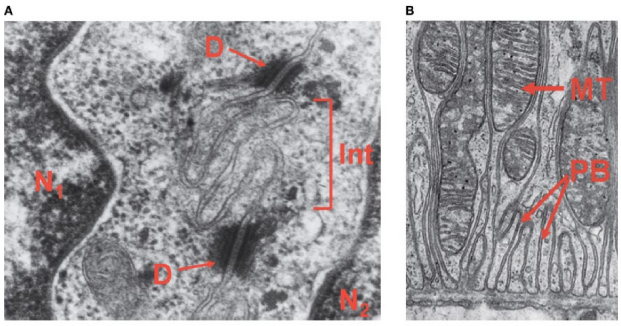


Figure 6.3. Microfoldings of the plasma membrane. A: Int: Lateral infolding. D: Desmosome. N1: Nucleus of cell 1. N2: Nucleus of cell 2. B: Basal infolding (PB). MT: Mitochondria.

Anchoring cell junctions

Cells adhere to one another and to the ECM through specific membrane molecules called cell adhesion molecules. These molecules are mostly transmembrane proteins which, together with other associated proteins, can be located in specific sites in the plasma membrane. The transmembrane proteins that participate in cell adhesion belong, mostly, to four families: cadherins, integrins, selectins and cellular adhesion molecules of the immunoglobulin superfamily (IgSF-CAM, immunoglobulin superfamily-cell adhesion molecules). These transmembrane proteins get attached to proteins of neighboring cells or the EMC (in the case of integrins). Proteins of these families can be monomeric (IgSF-CAM and selectins) or appear in a dimeric form; and so, cadherins form homodimers (dimers formed by the same type of protein), while integrins form heterodimers (dimers constituted by two different types of protein) **(table 6-1)**.

Table 6.1. Types of molecules involved in cell adhesion and types of interactions they establish with other molecules

Types of molecules	Homo-/heterophilic	Type of interaction	Transitory/stable
CADHERINS HOMODIMERS	HOMOPHILIC	CELL-CELL	STABLE JUNCTIONS (adhesion belts, desmosomes)
INTEGRINS HETERODIMERS α and β subunits	HETEROPHILIC with laminin/fibronectin	CELL-ECM	STABLE (hemidesmosomes or TRANSITORY (focal adhesions)
SELECTINS MONOMERS α and β subunits	HETEROPHILIC with sugars / IgSF-CAM	CELL-CELL	TRANSITORY
IgSF-CAM MONOMERS	HOMOPHILIC with IgSF-CAM or HETEROPHILIC with selectins or integrins	CELL-CELL	TRANSITORY

There is a large number of proteins associated to these transmembrane proteins, usually located on the cytoplasmic side of the membrane, participating in the structuring and stabilization of the junction and its anchoring to the cellular cytoskeleton. When an adhesion protein from one cell binds to another of the same type on a neighboring cell, we refer to this as **homophilic bonds**, while if it binds to different proteins (or other types of molecules), we refer to it as **heterophilic bonds**. These junctions can also be either of stable or temporary (v. table 6-1).

Cadherins

Cadherins constitute a family of proteins that contains Ca2+ binding sites **(fig. 6-4, A)**. Cadherins form homodimers and make Ca^{2+} dependent homophilic bonds. More than 40 different proteins belong to the cadherin family, among which we find **E-cadherin** (found in epithelial cells) and N-cadherin (present in nerve tissue cells, the lens of the eye, and skeletal and cardiac muscle cells).

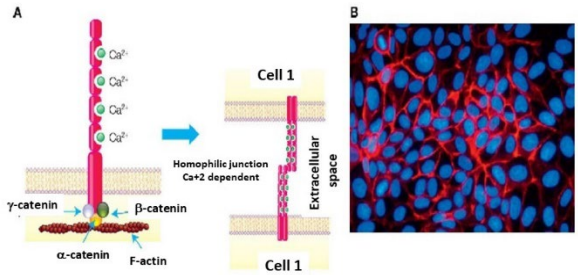


Figure 6.4. Structure of cadherins (A). Immunofluorescence image of E-cadherin (in red) in epithelial cells in culture.

Cadherins can be grouped in specific cell locations to form complex anchoring (or adhesion) structures. These structures are typically found in epithelial cells, although they may also be present in other tissues. There are two fundamental classes of complex anchoring structures that are cadherin dependent:

1. *Zonula adherens* or adhesion belt.
2. *Macula adherens* or desmosome.

We will now analyze the basic organization of each of these structures and some of the pathologies that arise when they get disorganized or undergo alterations.

ZONULA ADHERENS OR ADHESION BELT

The *zonula adherens* is a cellular belt with adhering function, normally found on the apical surface of the lateral cell membranes, which performs a vital role in the anchoring of epithelial cells **(fig. 6-5, A)**. **Cadherins** (belonging to the family of E-cadherins) are the transmembrane proteins that make up these junctions, whose cytoplasmic end joins to connecting proteins called α- and β-catenins **(fig. 6-5, B)**. The adhesion plate is bound to **actin filaments** by intermediary proteins. The actin filaments span through the cytoplasm, forming an intracytoplasmic anchoring network.

MACULA ADHERENS OR DESMOSOME

Macula adherens (also known as a desmosome) is 0.1-0.5 μm point shaped junction usually located in the lateral membranes of epithelial cells, providing anchorage between them **(fig. 6-5, C)**. We can also find desmosomes in intercalated disks that bind adjacent cardiomyocytes, where they play a key role in maintaining cardiac tissue structure. Desmosome forming transmembrane proteins are also cadherins (the so-called desmosomal cadherins: **desmocollins** and **desmogleins**). In this sense, desmosomes are structurally equivalent to adhesion belts, but they have a pointed morphology **(fig. 6-5, D)**. In desmosomes, the connecting proteins between the cadherins and the cytoskeleton form an intracellular plate made of **desmoplakin**, **plakophilin** and **plakoglobin**. This plate is directly attached to **intermediate filaments**, rather than to actin filaments. In the skin, intermediate filaments are **keratin (tonofilaments)**, **desmin** in the intercalated discs of cardiomyocytes and **vimentin** in other tissues.

There are different types of desmogleins, differently expressed depending on the tissue type. The main desmogleins in the epidermis of the skin (where desmosomes are abundant) are desmogleins 1 and 3. Two dermatological diseases related to desmogleins are **pemphigus vulgaris** and **pemphigus foliaceus**. These diseases are characterized by the presence of blisters on the cutaneous surface of patients. Pemphigus vulgaris (PV) is characterized by blisters that affect the surface of the skin, which burst easily **(fig. 6-6, A)**. These lesions can be found in the oral mucosa **(fig 6-6, B)** or extend as far as the skin of the face, neck, extremities and torso. Patients refer itchiness and pain in the affected areas, and they commonly have infections due to lesion ulceration. An anatomopathological exam of PV shows separation between keratinocytes in the skin, known as **acantholysis (fig. 6-6, C)**.

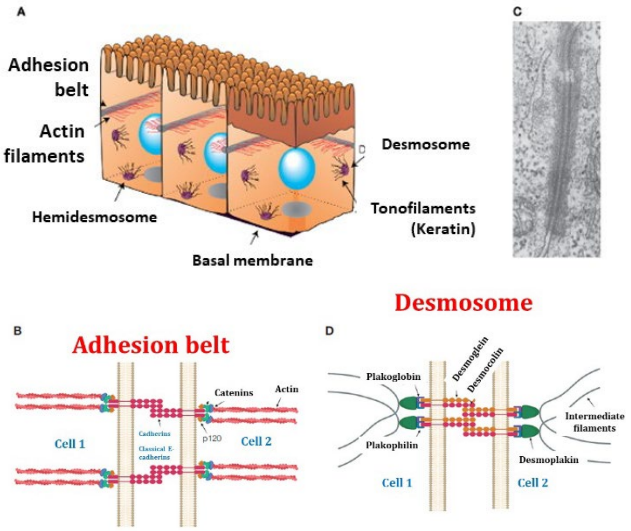


Figure 6.5. Structure of cell adhesion structures. A. Representation of desmosomes, adhesion belts and hemidesmosomes. B. Structure of an adhesion belt. C. Electron microscopy image of a desmosome. D. Structure of a desmosome.

These lesions are usually found above the **basal layer of the epidermis**. Immunofluorescence biopsy analysis reveals an **IgG deposit** on the keratinocyte surface, due to the autoimmune nature of the disease. Patient´s autoantibodies target **desmoglein 3** (mainly expressed on the basal layer of skin and mucosa), **desmoglein 1** (found mostly in the superficial layers of the skin and less commonly the mucosae) or both. Autoantibodies are responsible for desmosome deconstruction in **keratinocytes** (where they are very abundant) and blister appearance. This disease has a symptomatic treatment that involves corticosteroids and immunosuppressive drugs, as well as pain killers and antibiotics to control infections. **Pemphigus foliaceus** is also an autoimmune disease where an IgG deposit is observed targeting desmoglein 1. Therefore, the blisters will only be found in layers above the epidermis and will burst easily. The prognosis for this disease is better than that for PV, and its treatment consists of the use of systemic or intralesional corticosteroids.

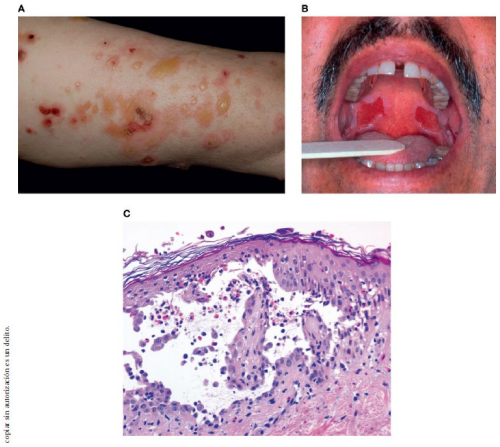


Figure 6.6. Macroscopic images of blisters (A) and oral lesions (B) in a patient with pemphigus vulgaris. C. Histological image of acantholysis of the skin.

Integrins

Integrins are a family of proteins that participate in the permanent or transient cell-EMC or cell-cell adhesion. They are heterodimers made up by two different subunits: **α subunit and β subunit (fig. 6-7, A).** The α-subunit has binding sites for divalent cations (Ca^{2+} and Mg^{2+}). These molecules bind to both ECM components and to intracellular cytoskeletal proteins by means of interaction with intermediary proteins that interact with **actin filaments**, but may also bind to intermediate **keratin filaments**. The extracellular domain of the β-subunit can bind to extracellular proteins (**laminin** and **fibronectin**). Integrins play an important role in the migration processes of embryonic and immune cells.

These proteins can be found as simple molecules or in groups forming complex adhesion structures, which constitute **hemidesmosomes** and **focal adhesions**. Hemidesmosomes are stable adhesion structures typically found in sessile cells, while focal adhesions are transient more commonly found in cells in motion. Focal adhesions are constantly undergoing a process of formation and destruction.

HEMIDESMOSOMES

Hemidesmosomes **(fig. 6-7, B)** are structures that, based on their morphology, would resemble half of a desmosome located in the basal part of the cell. These membrane differentiations allow anchoring of the cell to the extracellular basal membrane where the cell rests. However, the proteins that make up hemidesmosomes are different to those found in desmosomes, both the transmembrane anchoring proteins (in this case integrins instead of cadherins) and the adhesion plate proteins. The integrin that constitutes hemidesmosomes is the **α6β4-integrin**, and the proteins found in the plaque are known as **plakins**. Another transmembrane protein found in hemidesmosomes is **BPGA2** (bullous pemphigoid antigen-2). The plaque is connected to tonofilaments in the cytoplasmic side **(fig. 6-7, B)**. Hemidesmosomes increase the overall stability of epithelial tissues and allow them to settle in the ECM by binding of cytoskeletal tonofilaments to basal membrane components.

FOCAL ADHESIONS

Migratory cells bind to the ECM via focal adhesions **(fig. 6-7, C)**, formed by clusters of integrins interconnected with bundles of **actin filaments** by means of other intermediary proteins. These structures are very **dynamic**, in a process of constant formation and destruction during the process of cell migration, where they provide a connection at specific points with the substrate. They are abundant in **lamellipodia**, used by cells to travel through an inert substrate.

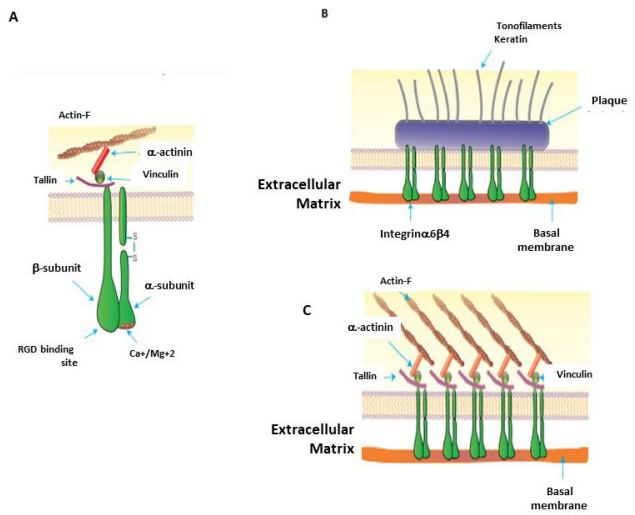


Figure 6.7. A. Structure of integrins. B. Representation of a hemidesmosome. C. Structure of focal adhesions.

Not only focal adhesions have an adhesion function, they also participate in intracellular signaling. Thus, integrin interaction with the substrate in focal adhesions results in phosphorylation of the **focal adhesion tyrosine kinase protein (FAK)**, a protein that promotes the activation of proteins involved in cellular migration and protects cells from anoikis. Integrins act as signal transductors, keeping the cell informed as to the conditions (mainly adhesion) of the extracellular surroundings. FAK hyperactivation has been associated with an increased metastatic capacity of tumor cells.

Integrins also mediate cell-cell contacts. These interactions are usually temporary and occur in the adhesion processes of immune system cells. They are highly involved in leukocyte extravasation process.

Selectins

Selectins are Ca^{2+} dependent adhesion molecules with affinity for **carbohydrates** found in other cells **(fig. 6-8, A)**. Therefore, they form heterophilic bonds (protein-carbohydrate). Rather than forming complex binding structures, they are involved in molecular contacts that are not visible under the microscope and are generally **temporary**. Similar to integrins, they frequently participate in the binding of leukocytes to endothelial cells, facilitating their exit from the blood vessels (extravasation) to an inflammatory site **(fig. 6-8, B)**.

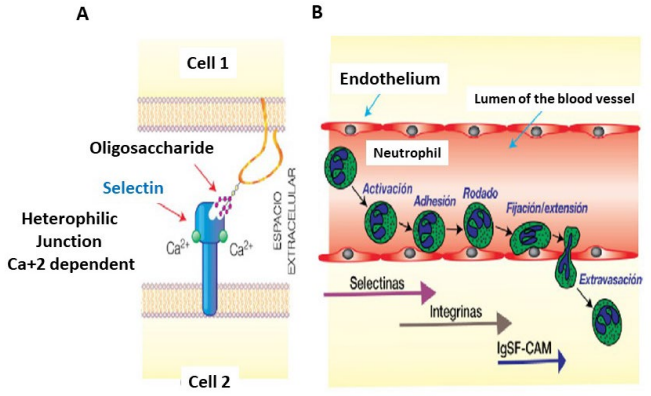


Figure 6.8. A. Structure of selectins. B. Participation of selectins (and other adhesion molecules, such as integrins) in the attachment of neutrophils to endothelial cells and extravasation to the sites of infection.

Leukocytes have **L-selectins** in their plasma membrane ('L' for leukocyte), able to interact with oligosaccharides in endothelial cells. Additionally, endothelial cells themselves have also selectins: **E-selectin** ('E' for "endothelium'), which in turn interact with oligosaccharides on the leukocyte surface. Selectins are involved in **inflammation** and **cancer**. In inflammatory processes, certain stimuli induce endothelial cells to produce selectins in their surface, thus facilitating the extravasation of leukocytes.

IgSF-CAM

The adhesion molecules of the IgSF-CAM immunoglobulin superfamily **(fig. 6-9)** possess bow-shaped domains, similar to those found in immunoglobulins. They form: a) heterophilic bonds when they interact with integrins or selectins; b) homophilic bonds, when they bind to other IgSF-CAM) **(fig.6-9)**. IgSF-CAMs have no Ca^{2+} binding sites and, therefore, they establish **Ca^{2+}-independent** interactions. They regulate cell junctions that are normally temporary. This protein superfamily is involved in adhesions of different cells, but we can stress their role in establishing contacts between cells of the immune system cells.

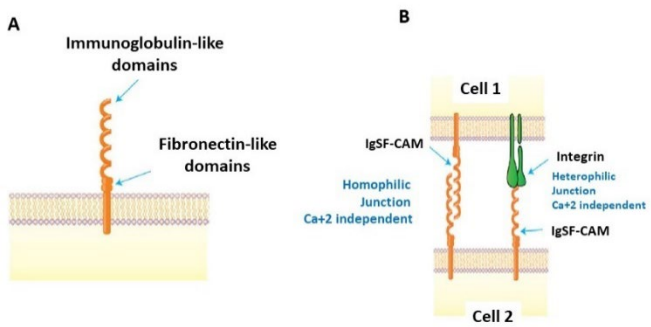


Figure 6.9. A. Structure of IgSF-CAM. B. Interaction between two molecules of IgSF-CAM, or between IgSF-CAM and selectins.

Occluding junctions (*Zonula occludens* or tight junction)

Occluding junctions (or 'tight junctions') are belt-shaped contacts, 1 μm in width, found almost **exclusively in epithelial cells** **(fig. 6-10)**. This **occlusive belt** is formed by a network of proteins that join the adjacent cell membranes, leaving the intercellular space occluded by a system of zipper-like structures. This can be seen, particularly, in the electron microscopy cryo-fracture images **(see fig. 6-10, C)**.

Tight junctions perform two essential roles:

1. They determine epithelial cells´ membrane **polarity**, as they separate the apical domain from the basolateral one, avoiding free lipid and protein diffusion between the two domains.
2. They **prevent substances from crossing** between epithelial cells (paracellular transport). However, sometimes cells can modify their tight junctions to allow the passage of certain substances, when these are in such excess that transport through the membrane is not sufficient to incorporate them. This is the case of amino acids and sugars. When they exceed the intestine´s ability to absorb them, they can be incorporated by paracellular transport.

The closure ridges are formed by interlocking rows of transmembrane proteins **(Fig. 6-10, B)**, particularly **occludins and claudins**, which bind directly to similar proteins in the adjoining cell.

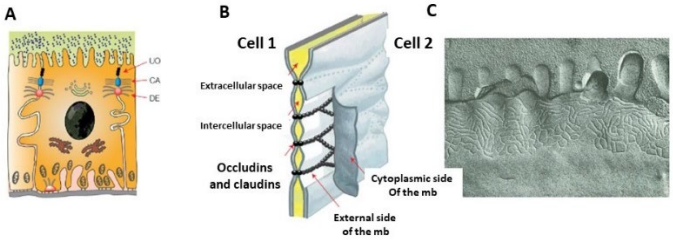


Figure 6.10. A. Structure of junctional complexes: UO: Occluding junction; CA: Adhesion belt; DE: Desmosome. B. Structure of occluding junctions. C. Electron microscopy image of an occluding junction.

Communicating junctions (gap, nexus or cleft junctions)

Communicating junctions are also called "**nexus**" or "**gap junctions**". They are macular (spot-like) that communicate the cytoplasm of neighboring cells, allowing direct transfer of ions and macromolecules **(fig 6-11)**. The basic unit in communicating junctions is the **connexon**. Cryo-fracture and negative contrast studies show that the connexons form a hexagonal grid. Each connexon is made up of six molecules of **connexin** (a transmembrane protein) around a central duct. The arrangement therefore creates a direct channel of communication (1.5-2 nm in diameter) between the cytoplasm of the two adjacent cells. Each nexus-type junction may contain a few or hundreds of connexons. These junctions allow movement of molecules up to 1.2 nm in diameter (such as inorganic ions, oligosaccharides, amino acids, nucleotides, vitamins and cAMP) between cells. The permeability of the channels is regulated by **Ca^{2+}** and **pH**: it decreases when pH decreases or when intracellular Ca^{2+} concentrations increase. When cells are damaged, there is a massive and instantaneous Ca^{2+} influx into the cell, so closure of these channels prevents any possible damage from being transmitted to neighboring cells.

This junction is responsible for the **electrochemical coupling** between adjacent cells, which is particularly relevant in the communication between some neurons and between cardiac muscular cells, where impulses must be conveyed quickly and synchronously. In the certain areas of the nervous system where there is a need for extremely rapid communication, there are numerous gap-junctions that constitute **electric synapses**. Another important example of gap junctions is the **ciliated epithelial cells** of the respiratory tract. The synchronous movement of the cilia to remove mucus from the airways requires rapid communication of signaling molecules through the cytoplasm of adjacent cells, which is achieved via gap junctions.

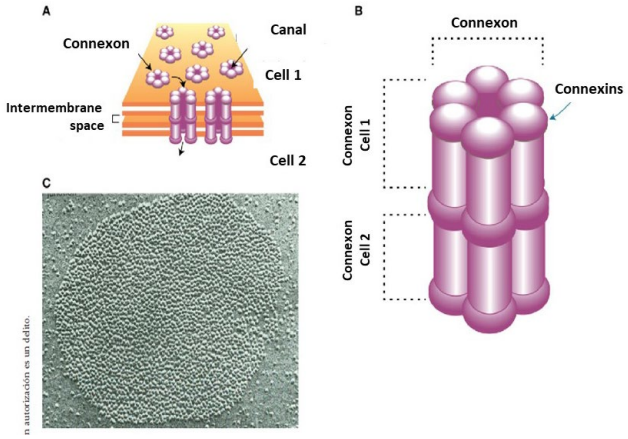


Figure 6.11. A. Structure of gap junctions. B. Structure of a connexon. C. Electron microscopy image of a gap junction.